

Euclidean Geometry In Mathematical Olympiads

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Problem 1.34



geometry

angle bisector

perpendicular bisector

Chapter 1

fgm

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Kagebaka

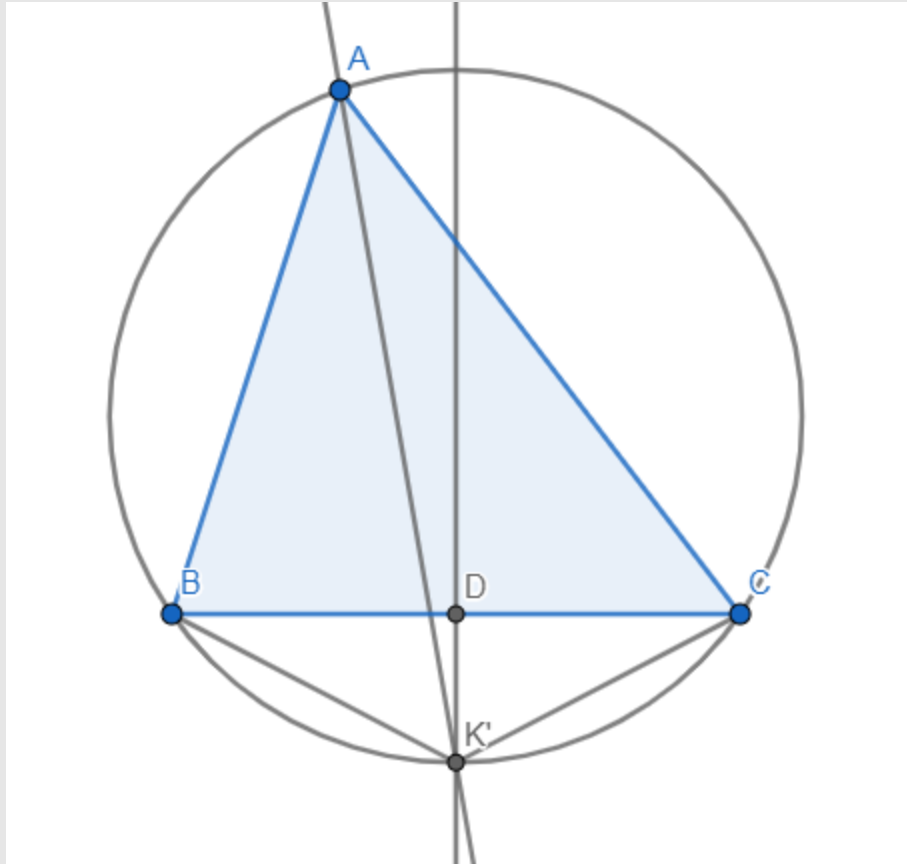
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#3

Solution

Let K' be the intersection of the perpendicular bisector of BC and (ABC) , and let D be the midpoint of BC :



Clearly, $K'DB \cong K'DC$ from LL, and since $ABCK'$ is cyclic, we have $\angle K'AB = \angle DCK'$ and $\angle K'AC = \angle DBK'$. However, since $\angle DBK' = \angle DCK'$, we then have $\angle K'AB = \angle K'AC$, so $K' = K$ and we're done. $\square$

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Can someone check this?

Slightly Different Solution